

STA201: Elements of Statistics & Probability

Assignment 3-5

Marks = 30

Question 1

The owner of a used car dealership is interested in researching how an automobile's age and selling price are related. Below is a sample of 12 used vehicles that the dealership sold during the course of the previous year. [6.0]

Age (years)	9	7	11	12	8	7	8	11	10	14	6	5
Price (thousand \$)	8.1	X-4	2.8	X-6	$\frac{X}{2}$	X	7.6	$\frac{X}{2}+3$	8.7	X-2	8.6	$X+\frac{1}{2}$

Here, X= 10

- Draw a scatter diagram and comment on the relation between the age of the car and its selling price. [1+1]
- Determine the Pearson correlation coefficient and the coefficient of determination and interpret it. [2.5+1.5]

Question 2

The iodine value is the amount of Iodine necessary to saturate a sample of 100 g of oil. In the table below the first row states Iodine and second row states oil respectively. [6.0]

132.0	129.0	120.0	113.2	105.0	92.0	84.0	83.2	88.4	59.0	80.0	81.5	71.0
46.0	48.0	51.0	52.1	54.0	52.0	59.0	58.7	61.6	64.0	61.4	54.6	58.8

- Determine the regression equation of 100 g of oil on the amount of Iodine. [2]
- Interpret the model. [1.5]
- What is the amount of oil when the amount of Iodine is 88 [1]
- Comment on the goodness of fit of the model. [1.5]

Question 3

Assume that the chances of a patient suffering from high blood pressure is 60%. It is also assumed that a course of meditation reduces the risk of high blood pressure by 45% and prescription of certain drugs reduces its chances by 55%. At a time, a patient can choose any one of the two options with equal probabilities. It is given that after going through one of the two options, the patient selected at random does not suffer from high blood pressure. Find the probability that the patient chose a course of meditation? [3.5]

Question 4

A discrete random variable X has the following probability mass function [6.0]

$$P(X = x) = \begin{cases} 2kx & x = 2, 4, 6 \\ k(x + 2) & x = 8 \\ 0 & \text{otherwise} \end{cases}$$

Where, k is a constant

- Show that $k = \frac{1}{34}$ [1]
- Find the exact value of $P(4 < x \leq 8)$ [1]
- Find the exact value of $P(2 < x < 4)$ [1]
- What is the expected value of the random variable X ? [1]
- What is the variance of the random variable X ? [1]
- Determine $Var(5 - 3X)$ [1]

Question 5

In each of 4 races, the Democrats have a 60% chance of winning. Assuming that the races are independent of each other, what is the probability that: [4.5]

- The Democrats will win 0 races, 1 race, 2 races, 3 races, or all 4 races? [2.5]
- The Democrats will win at least 1 race. [1.0]
- The Democrats will win a majority of the races. [1.0]

Question 6

Suppose on average, Nepal experiences 6 earthquakes per year. [4.0]

- What is the mean number of earthquakes in Nepal in the first four month of a year? [1.0]
- What is the probability that there'll be 7 earthquakes in Nepal in the next two years? [1.5]
- What is the probability that there'll be at least 9 earthquakes in Nepal in 2024? [1.5]